

README.en（英文完整版）

README.en.md

Ti Question Bank System (Competition Edition)

Project brief for the China College Computer Design Competition, Software Application & Development — Web Application track. Root overview: README.md · Chinese version: README.zh-CN.md

1. Project Overview

Ti is a single-repository full-stack system built for exam preparation, question-bank operations, and learning analytics.

Its verified implementation profile is summarized below:

Dimension	Description
Competition focus	Web application , built with Flask + Jinja templates + vanilla JS/CSS
Mobile extension	Native WeChat Mini Program using TypeScript + Less as a companion client sharing
Backend structure	Flask application factory + modular Blueprints
Data & infrastructure	Docker-based development with PostgreSQL 16 + Redis 7 + RQ Worker
API style	status / code / data / message envelope
Authentication	Session for Web, JWT Bearer Token for Mini Program / API
Default runtime	Docker development workflow driven by compose.dev.yml

This repository is not a split frontend-backend multi-repo setup, and it is not a SPA plus a separate API service. It is a unified Flask-centered business system serving both the Web interface and the Mini Program companion.

2. Competition Positioning and Problem Context

The project targets a combined scenario of question-bank management, practice workflows, learning feedback, and platform operations. Common issues in exam-preparation tools include:

- fragmented public banks and personal banks;
- disconnected records for favorites, mistakes, progress, and exams;
- inconsistent semantics between administration-oriented Web systems and mobile learning clients;
- missing operational support such as notifications, announcements, and community interaction.

Ti is positioned as follows:

- the Web application is the primary product surface, carrying public-bank discovery, bank details, analytics, forum interaction, and administration workflows;

- the WeChat Mini Program extends the same system to mobile learning scenarios without redefining the business model;
- a single repository keeps the data model, API semantics, and deployment workflow consistent.

That makes the project well aligned with the Web Application track: it is domain-oriented, full-stack, runnable, and verifiable.

3. Core Features

3.1 Learner-facing capabilities

Capability area	Implemented features	Related modules
Public banks	public bank plaza, bank detail pages, searching and categorized browsing	main page
Personal banks	create/join banks, sharing, publishing personal banks	user bank
Practice loop	question practice, answer recording, favorites, mistakes, progress, tags	quiz
Mock exams	exam creation, draft saving, submission, settlement records, templates	exam
Learning feedback	dashboards, subject/bank statistics, favorites and mistakes aggregation	main page
AI assistance	AI explanation for questions and asynchronous quiz handling	quiz handling

3.2 Platform and operation capabilities

Capability area	Implemented features	Related modules
Identity and profile	login, account binding, profile, sign-in, account setting	user
Notifications and popups	system notifications, announcements, popup notifications, popups	notification, popups
Community interaction	forum posts, comments, likes, chat messages	forum, chat
Administration	management of users, questions, subjects, notifications, configuration	admin
Coding extension	programming questions and online judging	coding

3.3 Shared backend semantics across clients

The project explicitly keeps the Web and Mini Program clients aligned on the same semantics, including at least:

- subjects;
- public banks;
- personal banks;
- questions;
- favorites and mistakes;
- user answers and learning progress;
- mock exams;
- notifications and profile data.

The Mini Program is therefore not an isolated side product, but a mobile entry point into the same business system.

4. System Architecture

```
flowchart LR
    web["Web Browser"] --> flask["Flask Web + API Layer"]
    mini["WeChat Mini Program"] --> flask
    flask --> db["PostgreSQL / SQLite"]
    flask --> redis["Redis"]
    worker["RQ Worker"] --> redis
    worker --> db
    worker --> ai["DashScope AI Explanation"]
```

4.1 Web layer

- The Web frontend is not a SPA. It uses Flask + Jinja templates + vanilla JS/CSS with server-side rendering.
- The shared shell lives in `app/modules/main/templates/main/shared/app_shell.html`, where layout, theming, dark/light mode, and common navigation are defined.
- Pages are organized under `app/modules/*/templates/`, with progressive enhancement through `static/css/` and `static/js/`.

4.2 Backend layer

- The application entry is `create_app()` in `app/__init__.py`.
- Business modules are registered centrally in `app/modules/__init__.py`, with 12 modules currently registered.
- The codebase uses an application factory, extension initialization, unified error handling, and a standardized response envelope.

4.3 Mini Program layer

- The Mini Program entry lives under `miniprogram-1/miniprogram/`.
- `app.json` declares both main-package and sub-package pages, with 58 pages declared in total.
- Requests are expected to flow through `utils/api-endpoints.ts` and `utils/api-client.ts` rather than scattered `wx.request` calls.

4.4 Data and task infrastructure

- Docker development defaults to PostgreSQL 16, as defined in `compose.dev.yml`.
- Redis is shared across caching, RQ queueing, and rate-limit storage.
- The worker container runs `rq worker -u $REDIS_URL $RQ_QUEUE_NAME` for background jobs.
- Health checks are available at `/api/ping` and `/api/ping?deep=1`.

5. Technical Highlights

5.1 Single-repo full stack with shared semantics

The Web client, Mini Program, Flask backend, database migrations, and Docker deployment files live in one repository. This reduces semantic drift and makes the engineering boundary easier to review, reproduce, and

demonstrate.

5.2 Web-primary design with a mobile companion

The project treats the Web application as the primary business surface and the Mini Program as a native mobile extension. This matches the competition track while still addressing real mobile usage scenarios.

5.3 Modular Flask architecture

The backend uses an application factory and Blueprint-based module registration. Responsibilities are split into modules such as auth , main , quiz , exam , user_bank , forum , and admin , which improves maintainability and feature isolation.

5.4 Dual authentication and compatibility strategy

- Web uses Session-based authentication;
- Mini Program / API uses JWT Bearer Token;
- Web XHR write requests must remain compatible with X-Requested-With and CSRF validation;
- API responses are being unified around status / code / data / message while remaining backward compatible.

5.5 Dockerized development and deployment workflow

The repository provides compose.dev.yml for local development with web , worker , postgres , redis , and backup services. Production is handled through compose.prod.yml together with Gunicorn and Nginx, improving repeatability for demos and deployment.

5.6 Verifiable engineering evidence

The project offers directly verifiable artifacts beyond UI pages:

- health endpoints: /api/ping , /api/ping?deep=1 ;
- Docker runtime entry: compose.dev.yml ;
- automated smoke tests: tests/test_smoke_health.py , tests/test_smoke_auth.py , tests/test_smoke_api.py .

6. Repository Structure

```
.
├── app/
│   ├── __init__.py
│   ├── core/
│   ├── models/
│   ├── modules/
│   ├── tasks/
│   └── miniprogram-1/
│       └── miniprogram/
│           ├── pages/
│           ├── packages/
│           ├── components/
│           └── utils/
├── static/
├── templates/
├── migrations/
└── # Flask app and business modules
    # create_app() factory
    # config, extensions, errors, utilities
    # ORM models
    # 12 business modules
    # background jobs
    # WeChat Mini Program project
    # main-package pages
    # sub-packages
    # shared components
    # API and utilities
    # Web static assets
    # global template directory
    # Alembic migrations
```

■■■ docker/	# Docker image resources
■■■ compose.dev.yml	# development compose file
■■■ compose.prod.yml	# production compose file
■■■ docs/	# supplementary docs
■■■ tests/	# pytest smoke tests

7. Docker Development and Validation

7.1 Default development path

```
cp .env.example .env
# fill in WECHAT_APPID / WECHAT_SECRET / DASHSCOPE_API_KEY as needed

docker compose --env-file .env -f compose.dev.yml up
```

Verified service facts in the development compose file:

- web : flask run --host 0.0.0.0 --port 8000 --reload
- worker : rq worker -u \$REDIS_URL \$RQ_QUEUE_NAME
- postgres : exposed on 5432:5432
- redis : shared storage for cache, rate limit, and queue
- backup : scheduled backup support in development mode

7.2 Database initialization

```
docker compose --env-file .env -f compose.dev.yml exec web flask db upgrade
```

7.3 Minimal runtime validation

```
# health checks
curl http://localhost:8000/api/ping
curl http://localhost:8000/api/ping?deep=1
```

Open the Web app in a browser: <http://localhost:8000>

7.4 Mini Program integration

- Open miniprogram-1/ with WeChat DevTools.
- API mode is controlled by miniprogram-1/miniprogram/utls/config.ts , which supports explicit prod and custom modes.
- Additional configuration details are documented in miniprogram-1/README_API_CONFIG.md .

8. Testing and Quality

The repository already includes pytest smoke tests for critical paths:

```
pytest
```

The existing checks cover:

- reachability of health endpoints;
- basic success and failure paths for login;
- accessibility and protection behavior for key pages and APIs.

Relevant test files:

- tests/test_smoke_health.py
- tests/test_smoke_auth.py
- tests/test_smoke_api.py

The project also contains several quality-oriented mechanisms at the engineering layer:

- global error handling and a unified response structure;
- Redis-backed rate-limit storage;
- logging with sensitive-data masking;
- optional Sentry integration in production.

Note: this section intentionally describes only what is already present in the repository and does not overstate coverage or production outcomes.

9. Additional Notes

9.1 Mini Program companion

Although the competition submission is Web-centered, the Mini Program is a native mobile extension sharing the same backend semantics. Relevant source files include:

- miniprogram-1/miniprogram/app.json
- miniprogram-1/miniprogram/utils/config.ts
- miniprogram-1/miniprogram/utils/api-client.ts
- miniprogram-1/miniprogram/utils/api-endpoints.ts

9.2 AI capability

Question explanation is integrated through the DashScope OpenAI-compatible interface, with configuration based on:

- DASHSCOPE_API_KEY
- DASHSCOPE_BASE_URL
- DASHSCOPE_MODEL
- DASHSCOPE_TIMEOUT

9.3 Supplementary documentation

- Development guide: docs/DEVELOPMENT.md
- Production guide: docs/PRODUCTION.md
- Docker command cheatsheet: docs/Docker命令速查.md
- Dual-machine sync guide: docs/Mac-Windows双机开发同步教程.md

If needed, this documentation can be extended with a presentation-oriented layer for screenshots, demo scripts, team roles, or defense notes.